4Pi-BRAINSPOT Toolbox for

Interferometric Ultra-High Resolution 3D Imaging through Brain Sections

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4Pi-BRAINSPOT Toolbox User Guide

1. GENERAL INFORMATION

4Pi-BRAINSPOT toolbox is distributed as accompanying software for manuscript: Interferometric Ultra-High Resolution 3D Imaging through Brain Sections. 4Pi-BRAINSPOT toolbox is developed for *in situ* point spread function (PSF) retrieval with the 4Pi single-molecule switching nanoscopy. It captures *in situ* interferometric information directly from the acquired interferometric single-molecule datasets, allowing *in-situ* 4Pi-PSF model that minimizes data-model disparity. The toolbox features an easy-to-use user interface including all steps of 4Pi reconstruction workflow from channel alignment, 4Pi-PSF segmentation, 4Pi *in situ* PSF retrieval, dynamic model update, 4Pi localization (supporting pupil-based 3D localization with cubic spline GPU implementation), drift correction, volume alignment, to super-resolution image reconstruction. It also contains a small single molecule dataset for users to run as an example.

1.1 Installation environment

- Windows 7 or later, 64 bit.
- MATLAB R2019b, 64 bit (downloadable at http://www.mathworks.com).
- CUDA 7.5 compatible graphics driver (downloadable at https://developer.nvidia.com/cuda-75-downloads-archive).

1.2 Installation of 4Pi-BRAINSPOT toolbox

1) Un-ZIP the ‘4Pi-BRAINSPOT toolbox.zip’ file including three folders: ‘4Pi-BRAINSPOT toolbox’, ‘Support’, and ‘Data_4Pi’.
2) Go to the ‘4Pi-BRAINSPOT toolbox’ folder and open the ‘main.m’ file.
3) Check the ‘support_path’ in the ‘main.m’ file to make sure that the path of the ‘Support’ folder is correct.
4) Run the ‘main.m’ file.

1.3 Update of 4Pi-BRAINSPOT toolbox

4Pi-BRAINSPOT toolbox for *in situ* 4Pi-PSF model estimation and 3D localization is available as **Supplementary software**. Further updates will be made freely available at Github repository.

2. STEP-BY-STEP GUIDE

2.1 User interface

The interface of 4Pi-BRAINSPOT toolbox includes eight modules: setup, data import, channel alignment, 4Pi-PSF segmentation, 4Pi *in situ* PSF retrieval, dynamic model update, 4Pi localization, and display modules (Figure 1). The brief introduction in each module is described as follows:

1) **Setup**: configure general setting parameters.
2) **Data import**: import the single molecule dataset from a 4Pi-SMSN configuration.
3) **Channel alignment**: calculate the position relationship between 4 detection channels and align their images into the same region of interest.
4) **4Pi-PSF segmentation**: segment interferometric emission pattern pairs from the 4-channel dataset to construct the 4Pi-PSF library.
5) **4Pi *in situ* PSF retrieval:** construct an *in situ* 3D PSF model directly from the acquired interferometric single-molecule datasets.
6) **Dynamic model update:** estimate cavity-phase-induced and objective misaligned interferometric aberration to form dynamic *in situ* model.
7) **4Pi localization**: reconstruct a 4Pi super-resolution image. This process includes segmentation, pupil-based 4Pi localization, rejection, 3D drift correction, and volume alignment.
8) **Display**: show the x-y view image of the reconstructed 3D volume with each molecule color-coded by its axial position.

The details in each module will be described in Sections 2.2 – 2.9.

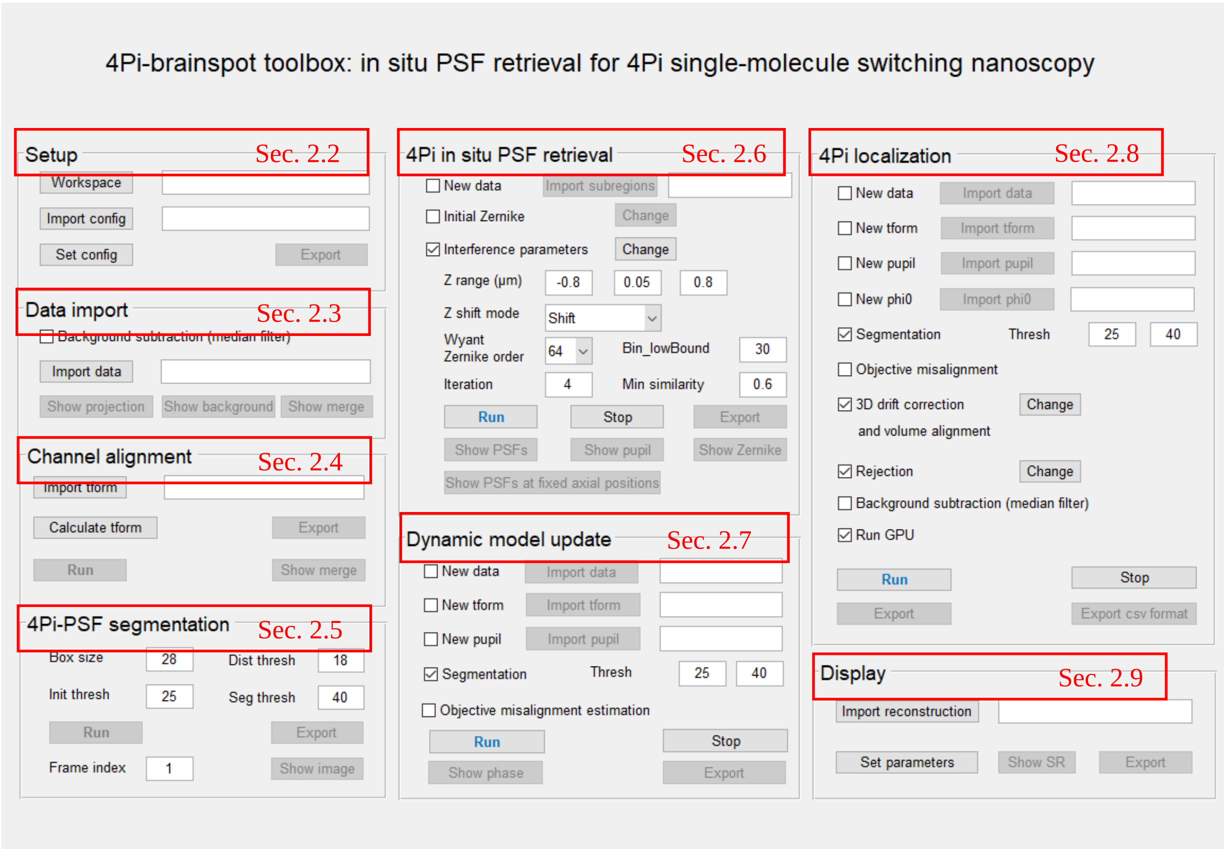

Figure 1. 4Pi-brainspot toolbox interface.

2.2 Setup module

Setup module is used to configure the workspace path and general setting parameters (Figure 2). The setting details in this module are described as follows:

- **'Workspace' button**: configure default input and output paths.
- **'Import config' button**: import general setting parameters.
- **'Set config' button**: modify setting parameters by users.
- **'Export' button**: export setting parameters.

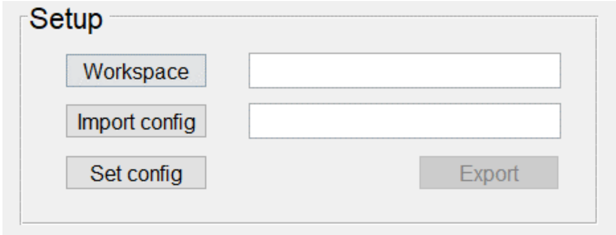

Figure 2. Setup module interface.

The setting parameters in ‘Set config’ button (Figure 3) include:

- **Pixel size**: effective pixel size on the camera.
- **Refractive index of immersion medium**: refractive index of the immersion medium of the objective lens.
- **Refractive index of sample medium**: refractive index of the imaging medium.
- **Lambda**: emission wavelength.
- **NA**: numerical aperture of the objective lens.
- **Camera offset**: offset on the camera.
- **Camera gain**: gain on the camera.
- **sCMOS camera model**: if this check box is selected, 4Pi-BRAINSPOT will carry out sCMOS calibration and enable ‘Import calibration file’ button. If not selected, 4Pi-BRAINSPOT will use EMCCD camera mode, which uses the same offset and gain for each pixel on the camera. By default, this option is not selected.
- **Import calibration file**: import sCMOS calibration file (e.g. ‘sCMOS_calibration_4Pi.mat’ file in ‘Data_4Pi’ folder).

Note: Protocols on how to characterize sCMOS camera’s pixel-dependent offset, gain, and variance are described in **Supplementary Method 2.3**.

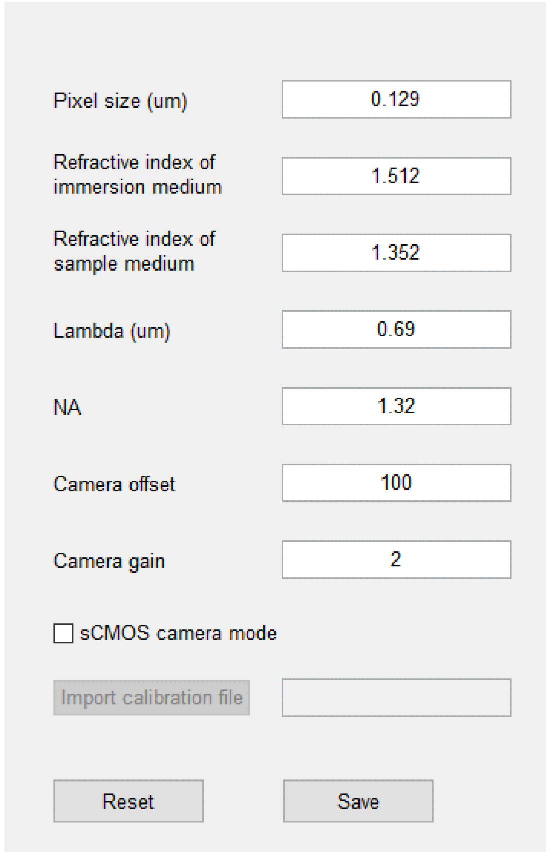

Figure 3. Setup parameters.

The workflow in setup module is as follows:

Step 1. Click 'Workspace' button and choose 'Data_4Pi' folder (described in Section 1.2).

Step 2. Click 'Set config' button and modify setup parameters by users (or use the default parameters). If the users want to choose sCMOS camera mode, select 'sCMOS camera mode' check box and import the sCMOS calibration file (e.g. 'sCMOS_calibration_4Pi.mat' file in 'Data_4Pi' folder).

Step 3. Click 'Export' button and save setup parameters.

Note: If you have a previously exported setup parameter file, you can click 'Import config' button to import this file (e.g. 'config_4Pi.mat' file in 'Data_4Pi' folder).

2.3 Data import module

Data import module is used to import the single molecule interferometric dataset from 4 detection channels in a 4Pi-SMSN configuration (Figure 4). Hereafter, the 4 detection channels are referred as *p1*, *s2*, *p2* and *s1*. The setting details in this module are described as follows:

- **‘Background subtraction’ check box**: Background subtraction option. If this box is selected, 4Pi-BRAINSPOT will estimate background using temporal median filtering, then carry out background subtraction. In this case, ‘Show background’ button will be enabled. By default, this option is not selected.
- **‘Import data’ button**: import the single molecule interferometric dataset.
- **‘Show projection’ button**: display the maximum-intensity projection images of the single molecule interferometric dataset.
- **‘Show background’ button**: display the background of the single molecule dataset.
- **‘Show merge’ button**: show the overlaid projection images from 4 channels compared to the *p1* channel (red in *p1* channel, green in selected channel).

Note:

- After the single molecule interferometric data is imported, ‘Show projection’ and ‘Show merge’ buttons will be enabled. Besides, ‘Show merge’ button in channel alignment module, as well as ‘Run’, ‘Export’ and ‘Show image’ buttons in 4Pi-PSF segmentation module will be disabled.
- 4Pi-BRAINSPOT supports the background subtraction option in cases with high background. During background subtraction, the statistical properties of the raw detected camera counts will be no longer maintained, it may decrease localization precisions.

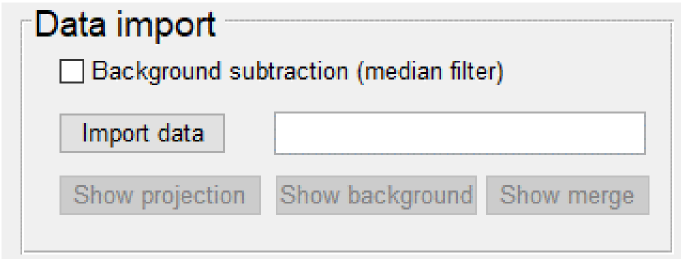

Figure 4. Data import module interface.

The workflow in data import module is as follows:

Step 1. Click ‘Import data’ button and choose ‘rawData_4Pi.mat’ file in ‘Data_4Pi’ folder.

Step 2. Click ‘Show projection’ button to show the maximum-intensity projection images of the single molecule interferometric dataset in a 4Pi-SMSN configuration (Figures 5A).

Step 3. Click ‘Show merge’ button to show the overlaid projection images from 4 detection channels (Figure 5B).

Note: The recommended image size is larger than 100 × 100 pixels.

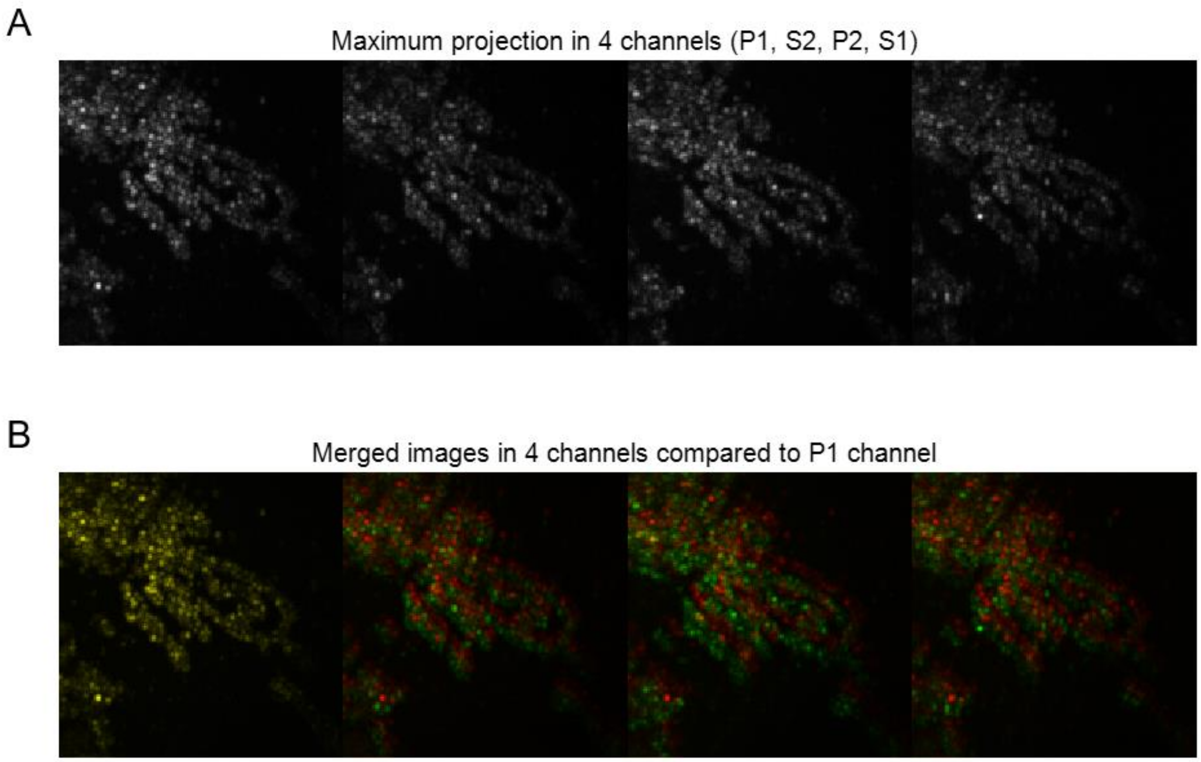

Figure 5. 4Pi single-molecule dataset. (A) Maximum projection images of single-molecule image sequences in 4 channels. (C) Overlaid projection images from P1 channel (red) and selected channels (green).

2.4 Channel alignment module

Channel alignment module is used to align the imported image stacks from 4 detection channels (Figure 6) into the same region of interest. The setting details in this module are described as follows:

- **'Import tform' button**: import the alignment calibration file (representing the position relationship between 4 detection channels) to align the images from the resting channels to *p1* channel.
- **'Calculate tform' button**: calculate the position relationship between 4 detection channels using affine transformation (as described in **Supplementary Method 3.2**).
- **'Export' button**: export the alignment calibration file.
- **'Run' button**: align the images from the resting channel to *p1* channel based on the alignment calibration file.
- **'Show merge' button**: show the overlaid projection images from 4 channels after alignment calibration.

Note:

- After the position relationship between 4 detection channels is imported or calculated, 'Export' and 'Run' buttons will be enabled.
- After clicking 'Run' button to align the images from the resting channels to *p1* channel, 'Show merge' button in channel alignment module and 'Run' button in 4Pi-PSF segmentation module will be enabled.

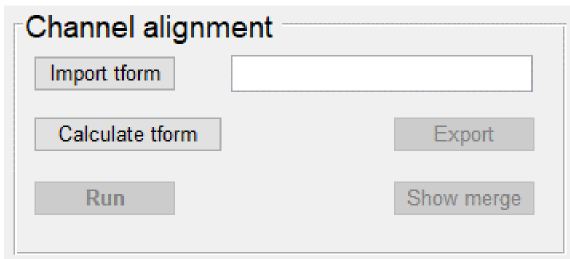

Figure 6. Channel alignment module interface.

The workflow in channel alignment module is as follows:

Step 1. Click 'Calculate tform' button to calculate the position relationship between 4 detection channels.

Step 2. Click 'Export' button and save the alignment calibration file.

Step 3. Click 'Run' button to align the images from the resting channels to *p1* channel.

Step 4. Click 'Show merge' button to show the overlaid projection images from 4 detection channels after alignment calibration (Figure 7).

Note: If you have a previously generated alignment calibration file, you can click 'Import tform' button to import this file (e.g. 'tform_all.mat' file in 'Data_4Pi' folder).

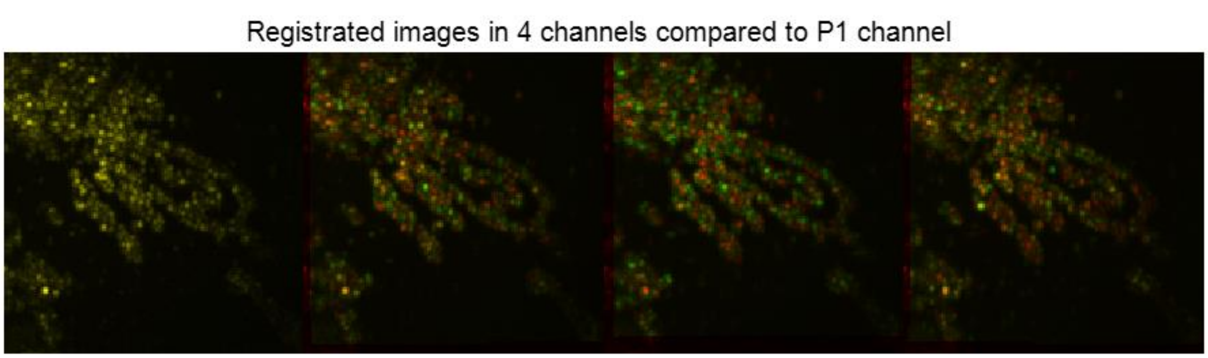

Figure 7. Overlaid projection images from 4 channels after alignment calibration.

2.5 4Pi-PSF segmentation module

4Pi-PSF segmentation module is used to crop pairs of sub-regions containing single molecules in a 4Pi-SMSN configuration (Figure 8). The setting details in this module are described as follows (explanations for setting parameters are described in **Supplementary Method 3.2**):

- **Box size**: sub-region size of the cropped sub-regions from the single molecule interferometric dataset (unit: pixel). Recommended value is from 28 to 40.
- **Dist thresh**: distance threshold to make sure that each selected sub-region contains only one molecule (unit: pixel). Recommend value is around Box size divide by sqrt(2).
- **Init thresh**: initial intensity threshold to obtain the candidate sub-regions.
- **Seg thresh**: segmentation threshold to select sub-regions with higher photon counts compared to initial candidate sub-regions.
- **'Run' button**: crop sub-regions from the dataset.
- **'Export' button:** export the cropped sub-regions.
- **Frame index:** frame index in the single-molecule interferometric dataset.
- **'Show image' button**: show the cropped sub-regions with the given frame index.

Note: After the segmentation process with 'Run' button is completed, 'Export' and 'Show image' buttons will be enabled.

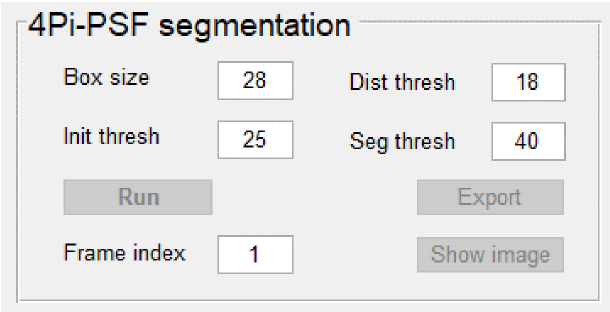

Figure 8. 4Pi-PSF segmentation module interface.

The workflow in 4Pi-PSF segmentation module is as follows:

Step 1. Configure setting parameters including 'Box size', 'Dist thresh', 'Init thresh', and 'Seg thresh'.

Step 2. Click 'Run' button to crop sub-regions.

Step 3. Click 'Export' button and save the cropped sub-regions.

Step 4. Set 'Frame index'.

Step 5. Click 'Show image' button to show the cropped sub-regions in 4 channels (Figure 9).

Note:

- After segmentation is finished, the user can set ‘Frame index’ to show the cropped sub-regions in the single molecule interferometric dataset with green boxes.
- The user can adjust ‘Dist thresh’, ‘Init thresh’, and ‘Seg thresh’ parameters to make sure that the 4Pi-PSF library has enough number of selected sub-regions (more than 2000 sub-regions are recommended) and each selected sub-region only contains one molecule with enough brightness. In cases with high background, the user can increase ‘Init thresh’ and ‘Seg thresh’ simultaneously to reduce the influence of background. In cases with high-density molecules, the user can increase ‘Dist thresh’ to get rid of overlapped molecules. In cases with non-uniform illumination, the user can increase ‘Seg thresh’ to select brighter molecules. In cases with few detected molecules, the user can decrease ‘Dist thresh’, ‘Init thresh’, and ‘Seg thresh’, but some PSFs with low quality may be included.

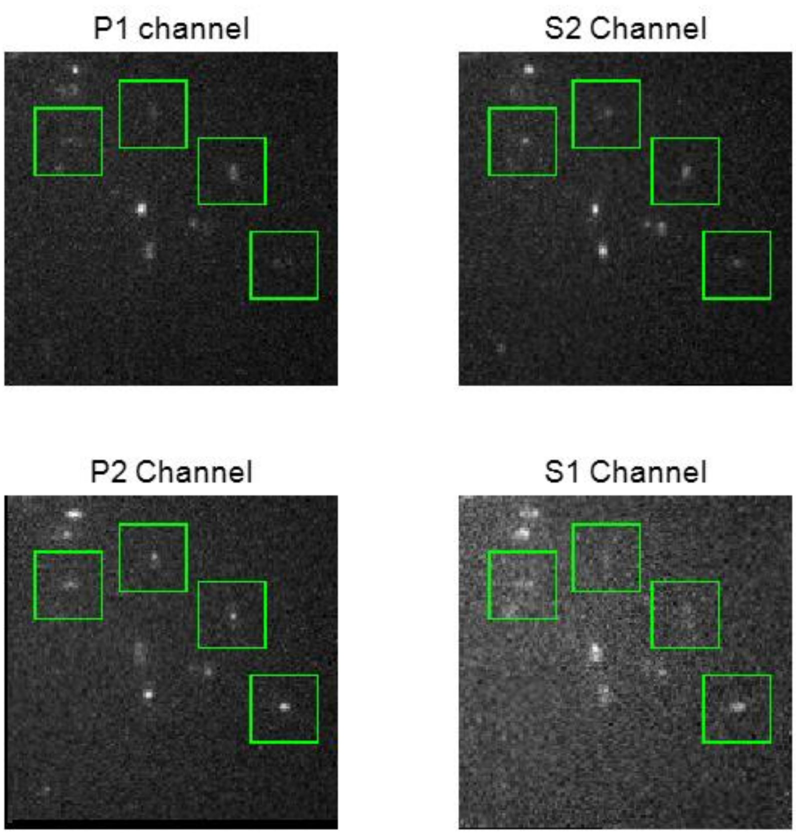

Figure 9. Cropped sub-regions in 4 channels.

2.6 4Pi *in situ* PSF retrieval module

4Pi *in situ* PSF retrieval module is used to generate an *in situ* 4Pi-PSF model directly from the cropped sub-regions of the single molecule interferometric dataset (Figure 10). 4Pi-BRAINSPOT starts with an ideal PSF (i.e. with constant pair of pupils) and then assigns each cropped sub-region to a temporary axial position through cross correlation with this ideal template. These axially assigned sub-regions are subsequently grouped, aligned, and averaged to form a 3D PSF stack which is used to retrieve a new pair of pupil estimation from the bottom and top objectives through coherent 4Pi phase retrieval. This new pair of pupils are then used to generate an updated template. This process iterates until good agreement between the cropped sub-regions and the retrieved model is reached. The setting details in this module are described as follows (additional explanations on how to optimize parameters are described in **Supplementary Method 3.1**):

- **'New data' check box**: if this box is selected, 'Import subregion' button will be enabled, and the previous sub-regions can be imported. If not selected, 4Pi-BRAINSPOT uses the sub-regions from 4Pi-PSF segmentation module. By default, this option is not selected.
- **'Import subregion' button**: import the sub-regions by users.
- **'Initial Zernike' check box**: if this box is selected, the users can modify the initial coefficients of 21 Zernike modes from the bottom and top objectives (Wyant order, from vertical astigmatism to tertiary spherical aberration, unit: $\lambda/2\pi$). By default, this option is not selected.
- **'Change' button**: modify the initial coefficients of 21 Zernike modes.
- **'Interference parameters' check box**: if this box is selected, the users can modify the interference parameters in 4Pi *in situ* PSF retrieval. By default, this option is selected.
- **'Change' button**: modify the interference parameters.
- **Z range**: three input texts from left to right are minimum axial position, axial step size, and maximum axial position.
- **'Z shift mode' popup menu**: lateral and axial positions optimization mode for averaged sub-regions. (1) Z shift mode = 'Shift', meaning the lateral and axial positions are optimized. (2) Z shift mode = 'No shift', meaning the lateral and axial positions are not optimized. The default is set to 'Shift'.
- **Wyant Zernike order**: number of output Zernike modes (Wyant order).
- **Bin_lowBound**: number threshold to reject an axial position group which contains fewer sub-regions than this threshold.
- **Iteration**: iteration number of 4Pi *in situ* PSF retrieval.

- **Min similarity**: similarity threshold to reject a sub-region with similarity lower than this threshold (from 0 to 1).
- **'Run' button**: carry out *in situ* 3D PSF model generation.
- **'Stop' button**: stop *in situ* 3D PSF model generation.
- **'Export' button**: export the *in situ* 3D PSF model.
- **'Show PSFs' button**: show the retrieved 4Pi-PSFs along the axial direction.
- **'Show pupil' button**: show the retrieved pupil (including its magnitude and phase).
- **'Show Zernike' button**: show the decomposed Zernike coefficients from the retrieved phase.
- **'pupil_showPSFs_fixed_pushbutton' button:** show the retrieved 4Pi-PSFs at the fixed axial direction.

Note:

- After an *in situ* 3D PSF model is generated, 'Export', 'Show PSFs', 'Show pupil', and 'Show Zernike' buttons will be enabled.
- Z range is usually larger than the thickness of sample to cover the all detected axial range of molecules.

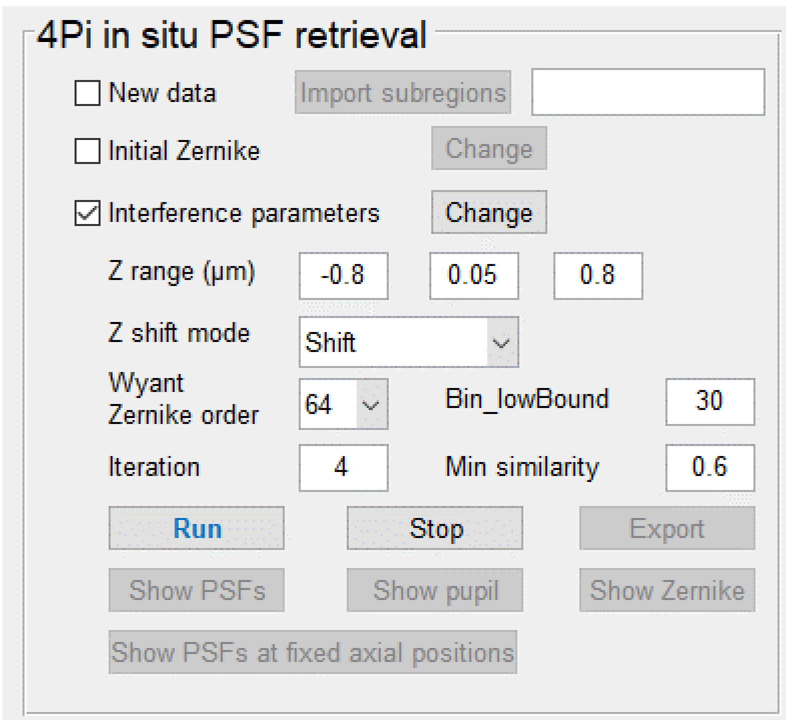

Figure 10. 4Pi *in situ* PSF retrieval module interface.

The setting parameters in Interference parameters (Figure 11) include:

- **Phase difference:** phase difference between s- and p-polarized fluorescence controlled to be around -π/2.
- **Transmission ratio:** transmission efficiency ratio between upper and lower paths.
- **Cavity phase:** phase difference between the two optical paths when the emitter is at the coincided focal plane.
- **Coherence strength:** wavelength-dependent coherence factor between 0 and 1.
- **Min photon:** photon threshold to reject single molecules with photon counts lower than this value.
- **Blur parameters:** optical transfer function (OTF) rescaling of the detected fluorophore.

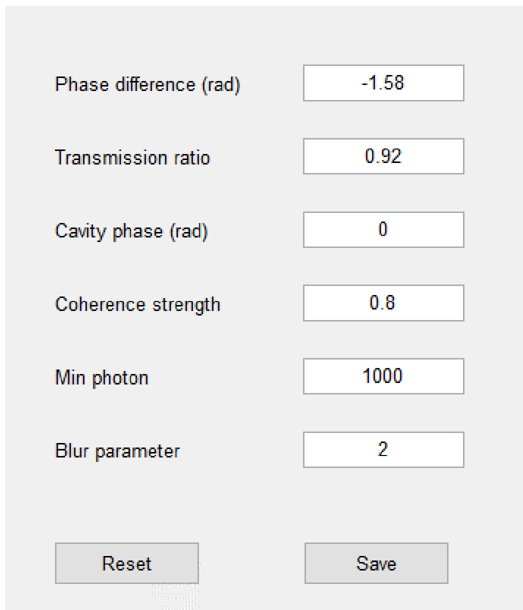

Figure 11. Setting parameters in Interference parameters.

The workflow in 4Pi *in situ* PSF retrieval module is as follows:

Step 1. Configure setting parameters including ‘Initial Zernike’, ‘Interference parameters’, ‘Z range’, ‘Z shift mode’, ‘Wyant Zernike order’, ‘Bin_lowBound’, ‘Iteration’, and ‘Min similarity’.

Step 2. Click ‘Run’ button to generate an *in situ* 3D PSF model.

Step 3. Click ‘Export’ button and save the generated *in situ* 3D PSF model.

Step 4. Click ‘Show PSFs’ button to show the retrieved PSFs along the axial direction (Figure 12).

Step 5. Click ‘Show pupil’ button to show the retrieved pupil.

Step 6. Click ‘Show Zernike’ button to show the decomposed Zernike coefficients from the retrieved phase.

Step 7. Click ‘Show PSFs at fixed axial positions’ to show the retrieved PSFs at the fixed axial direction.

Note:

- If you have a previously generated sub-region file, you can select ‘New data’ check box and click ‘Import sub-regions’ button to import this file (e.g. ‘subregion_chs.mat’ file in ‘Data_4pi’ folder).
- If you want to stop 4Pi *in situ* PSF retrieval, you can click ‘Stop’ button, which requires a few minutes to stop the process.
- The target of 4Pi-BRAINSPOT is to deal with isolated cells and native tissue specimens. If the specimen is very thin, the range of localization may not be enough for reliable model generation.
- Due to the phase wrapping, the decomposed Zernike value may not be reliable. In this case, we focus PSF shape information and ignore the Zernike fitting results.

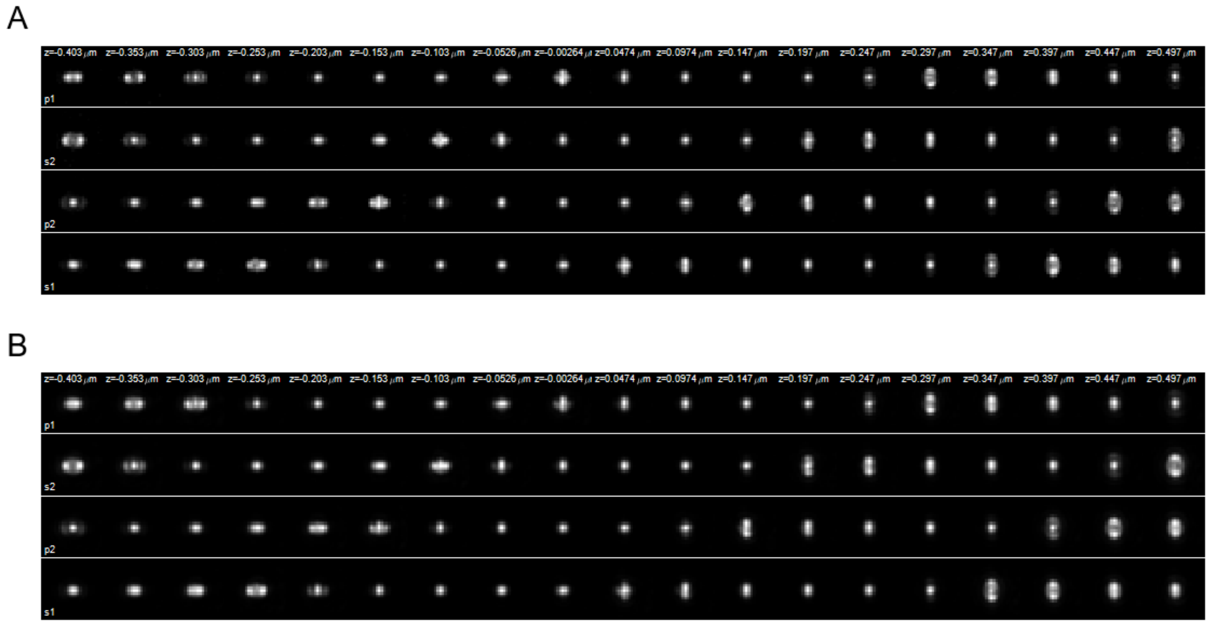

Figure 12. x-y views of measured z-group 4Pi-PSFs (A) and retrieved 4Pi-PSFs model using 4Pi-brainspot (B).

2.7 Dynamic model update module

Dynamic model update module is used to estimate interferometric aberrations caused by changes in the cavity phase and misalignment of the objective (Figure 13). These aberrations can dynamically vary across different time windows. The setting details in this module are described as follows (explanations for setting parameters are described in **Supplementary Method 3.5 and 3.6**):

- **'New data' check box**: if this box is selected, 'Import data' button will be enabled, and the single molecule interferometric dataset can be imported. If not selected, 4Pi-BRAINSPOT uses the dataset from data import module. By default, this option is not selected.
- **'Import data' button**: import the single molecule interferometric dataset by users (described in Section 2.2 'data import module').
- **'New tform' check box**: if this box is selected, 'Import tform' button will be enabled, and the alignment calibration file can be imported. If not selected, 4Pi-BRAINSPOT toolbox uses the calibration file from channel alignment module. By default, this option is not selected.
- **'Import tform' button**: import the alignment calibration file by users (described in Section 2.3 'Channel alignment module').
- **'New pupil' check box**: if this box is selected, 'Import pupil' button will be enabled, and the *in situ* 3D PSF model can be imported. If not selected, 4Pi-BRAINSPOT toolbox uses the *in situ* 3D PSF model from 4Pi *in situ* PSF retrieval module. By default, this option is not selected.

- **‘Import pupil’ button**: import the *in situ* 3D PSF model by users (described in Section 2.6 ‘4Pi *in situ* PSF retrieval module’).
- **‘Segmentation’ check box**: if this box is selected, the threshold can be set to crop sub-regions. By default, this option is selected.
- **Thresh:** initial and segmentation thresholds (described in Section 2.5 ‘4Pi-PSF segmentation module’).
- **‘Objective misalignment’ check box**: if this box is not selected, the dynamic *in situ* model will consider cavity phase. If this box is selected, the model will consider both cavity-phase-induced and objective-misaligned interferometric aberration. By default, this option is not selected.
- **‘Run’ button**: carry out cavity phase and objective misalignment estimation.
- **‘Stop’ button**: stop the model estimation.
- **‘Show phase’ button**: show the estimated cavity phase and objective misalignment.
- **‘Export’ button**: export the model estimation.

Notes:

- For importing data, 4Pi-BRAINSPOT allows to import multiple datasets.
- For importing pupils, 4Pi-BRAINSPOT allows importing multiple pupils for multi-section imaging, where the corresponding pupil is used in each optical section. These multiple pupils will be used for multiple sections in the order of their names. If you want to use one single pupil for multiple sections, you need to copy this pupil for multiple times and then import them together.
- For segmentation, the threshold selection is described in Section 2.5 ‘4Pi-PSF segmentation module’.

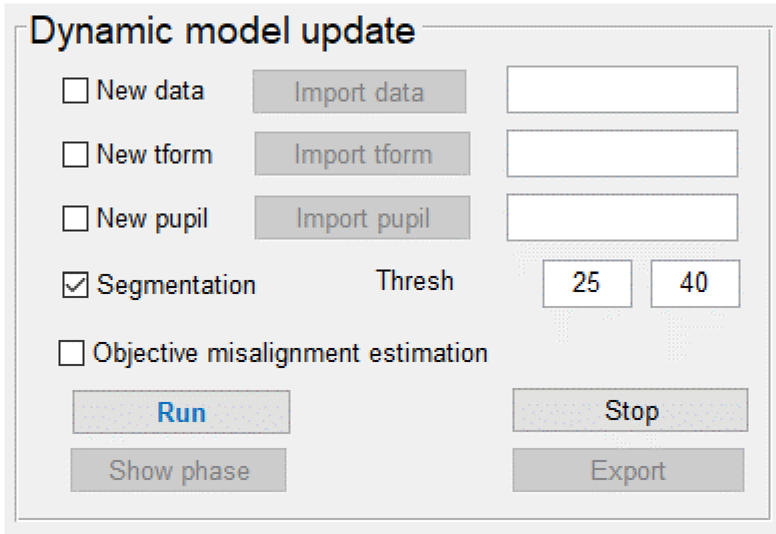

Figure 13. Dynamic model module interface.

The workflow in cavity phase and objective misalignment module is as follows:

Step 1. Configure 'Thresh' parameters in segmentation (related to initial and segmentation threshold in Section 2.5).

Step 2. Click 'Objective misalignment' check box to indicate whether to include objective misalignment estimation.

Step 3. Click 'Run' button to carry out cavity phase and objective misalignment estimation.

Step 4. Click 'Show phase' button to show the estimated results.

Step 5. Click 'Export' button and export the estimated results.

Note:

- If you have previously generated single-molecule interferometric datasets, you can select 'New data' check box and click 'Import data' button to import one or multiple files (e.g. 'rawData_4Pi.mat' file in 'Data_4pi' folder).
- If you have a previously generated alignment calibration file, you can select 'New tform' check box and click 'Import tform' button to import this file (e.g. 'tform_all.mat' file in 'Data_4pi' folder).
- If you have a previously generated *in situ* 4Pi-PSF model, you can select 'New pupil' check box and click 'Import pupil' button to import this file (e.g. 'probj.mat' file in 'Data_4pi' folder).
- If you want to stop the estimation, you can click 'Stop' button.

2.8 4Pi localization module

4Pi localization module is used to construct a 3D super-resolution image from single molecule interferometric dataset (Figure 14). This module includes segmentation, dynamic *in situ* model incorporating cavity phase and objective misalignment, 4Pi pupil-based 3D localization (supporting GPU version), rejection, 3D drift correction, and volume alignment. The setting details in this module are described as follows (explanations for setting parameters are described in **Supplementary Method 3.7**):

- **'New data' check box**: if this box is selected, 'Import data' button will be enabled, and the single molecule interferometric dataset can be imported. If not selected, 4Pi-BRAINSPOT uses the dataset from data import module. By default, this option is not selected.
- **'Import data' button**: import the single molecule interferometric dataset by users (described in Section 2.2 'data import module').
- **'New tform' check box**: if this box is selected, 'Import tform' button will be enabled, and the alignment calibration file can be imported. If not selected, 4Pi-BRAINSPOT toolbox uses the calibration file from channel alignment module. By default, this option is not selected.
- **'Import tform' button**: import the alignment calibration file by users (described in Section 2.3 'Channel alignment module').
- **'New pupil' check box**: if this box is selected, 'Import pupil' button will be enabled, and the *in situ* 3D PSF model can be imported. If not selected, 4Pi-BRAINSPOT toolbox uses the *in situ* 3D PSF model from 4Pi *in situ* PSF retrieval module. By default, this option is not selected.
- **'Import pupil' button**: import the *in situ* 3D PSF model by users (described in Section 2.6 '4Pi *in situ* PSF retrieval module').
- **'New phi0' check box**: if this box is selected, 'Import phi0' button will be enabled, and the cavity phase and objective misalignment estimation can be imported. If not selected, 4Pi-BRAINSPOT toolbox uses the cavity phase and objective misalignment estimation from cavity phase and objective misalignment module. By default, this option is not selected.
- **'Import phi0' button**: import the cavity phase and objective misalignment estimation by users (described in Section 2.7 'cavity phase and objective misalignment module').
- **'Segmentation' check box**: if this box is selected, the threshold can be set to crop sub-regions. By default, this option is selected.
- **Thresh:** initial and segmentation thresholds (described in Section 2.5 '4Pi-PSF segmentation module').

- **‘Objective misalignment’ check box**: if this box is selected, the dynamic *in situ* model will consider objective-misaligned interferometric aberration. By default, this option is not selected.
- **‘3D drift correction and volume alignment’ check box**: if this box is selected, 4Pi-BRAINSPOT will carry out 3D drift correction and volume alignment, and the corresponding setting parameters in this process can be modified. By default, this option is selected.
- **‘Rejection’ check box**: if this box is selected, 4Pi-BRAINSPOT will carry out the rejection process, and the corresponding setting parameters in this process can be modified. By default, this option is selected.
- **‘Background subtraction’ check box**: Background subtraction option. If this box is selected, 4Pi-BRAINSPOT will estimate background using temporal median filtering, then carry out background subtraction. By default, this option is not selected.
- **‘Run GPU’ check box**: 4Pi-BRAINSPOT currently only supports the GPU version for pupil-based 4Pi localization. Please check CUDA compatible graphics driver (described in Section 1.1 ‘Installation environment’).
- **‘Run’ button**: carry out 3D super-resolution reconstruction.
- **‘Stop’ button**: stop 3D super-resolution reconstruction.
- **‘Export’ button**: export 3D super-resolution reconstruction results.
- **‘Export csv format’ button**: export (x, y, z) positions of single molecules by using ‘csv’ format.

Notes:

- For importing data, 4Pi-BRAINSPOT allows to import multiple datasets.
- For importing pupils, 4Pi-BRAINSPOT allows importing multiple pupils for multi-section imaging, where the corresponding pupil is used in each optical section. These multiple pupils will be used for multiple sections in the order of their names. If you want to use one single pupil for multiple sections, you need to copy this pupil for multiple times and then import them together.
- For segmentation, 4Pi-BRAINSPOT transforms the identified sub-region centers from *p1* channel to the resting channels, and crops pars of sub-regions from 4 detection channels. The sub-region size of the cropped sub-regions is set to 16×16 pixels, and the distance threshold is set to 10 when the image size is larger than 100×100 pixels, otherwise this threshold is set to 6. The initial and segmentation thresholds should be the same with the thresholds described in Section 2.5 ‘4Pi-PSF segmentation module’.
- For 4Pi localization, 4Pi-BRAINSPOT generates channel-specific PSF models to avoid imaging artifacts and localization imprecisions.

- 4Pi-BRAINSPOT supports GPU version for pupil-based 4Pi localization. Please check GPU environment (described in Section 1.1 'Installation environment') before running the code.
- 4Pi-BRAINSPOT supports the background subtraction option in cases with high background. During background subtraction, the statistical properties of the raw detected camera counts will be no longer maintained, it may decrease localization precisions.

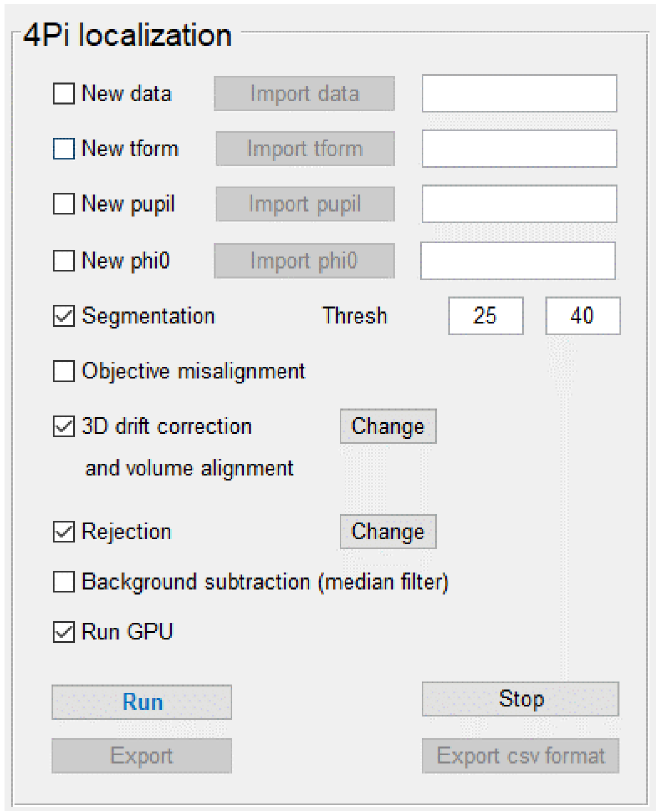

Figure 14. 4Pi localization module interface.

The setting parameters in 3D drift correction and volume alignment (Figure 15) include:

- **Frame bin**: number of frames used to construct individual 3D volumes for 3D drift correction.
- **Initial Z offset**: axial position offset to avoid negative z positions during 3D volume reconstruction.
- **Step interval**: axial step size for multi-section imaging.

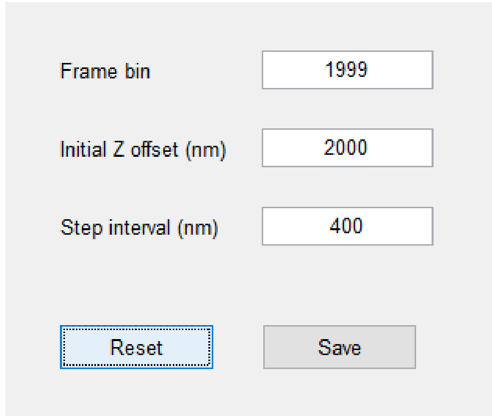

Figure 15. Setting parameters in 3D drift correction and volume alignment.

The setting parameters in rejection process (Figure 16) include:

- **Min photon**: photon threshold to reject single molecules with photon counts lower than this value.
- **LLR threshold**: log-likelihood ratio (LLR) threshold to reject single molecules with LLR higher than this value.
- **Max xy uncertainty**: localization uncertainty in the lateral dimension to reject single molecules with uncertainty higher than this value.
- **Max Z uncertainty**: localization uncertainty in the z dimension to reject single molecules with uncertainty higher than this value.
- **Z mask**: axial position mask to reject single molecules with axial positions beyond this range.

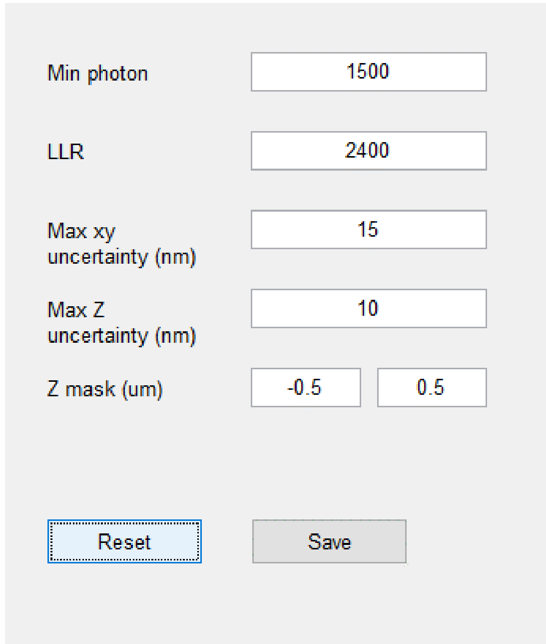

Figure 16. Setting parameters in rejection process.

The workflow in 4Pi localization module is as follows:

Step 1. Configure ‘Thresh’ parameters in segmentation (related to initial and segmentation threshold in Section 2.5).

Step 2. Configure setting parameters including ‘Objective misalignment’, ‘3D drift correction and volume alignment’, ‘Rejection’, and ‘Background subtraction’.

Step 3. Click ‘Run’ button to carry out 4Pi localization.

Step 4. Click ‘Export’ button and export 3D super-resolution reconstruction results.

Step 5. Click ‘Export csv format’ button and export (*x*, *y*, *z*) positions of single molecules by using ‘csv’ format.

Note:

- If you have previously generated single-molecule interferometric datasets, you can select ‘New data’ check box and click ‘Import data’ button to import one or multiple files (e.g. ‘rawData_4Pi.mat’ file in ‘Data_4pi’ folder).
- If you have a previously generated alignment calibration file, you can select ‘New tform’ check box and click ‘Import tform’ button to import this file (e.g. ‘tform_all.mat’ file in ‘Data_4pi’ folder).

- If you have a previously generated *in situ* 3D PSF model, you can select ‘New pupil’ check box and click ‘Import pupil’ button to import this file (e.g. ‘probj.mat’ file in ‘Data_4pi’ folder).
- If you have a previously estimated parameters for dynamic model, you can select ‘New Phi0’ check box and click ‘Import Phi0’ button to import this file (e.g. ‘phi0_est.mat’ file in ‘Data_4pi’ folder).
- If you want to stop 4Pi localization, you can click ‘Stop’ button.
- GPU version for pupil-based 4Pi localization require GPU compatible graphics driver, please check GPU environment before running the code.

2.9 Display module

Display module is used to generate the x-y view of the reconstructed super-resolution image with each molecule color-coded by its axial position (Figure 17). The setting details are listed as follows:

- ‘Import reconstruction’ button: import the 3D super-resolution reconstruction result by users (described in Section 2.8 ‘4Pi localization module’).
- ‘Set parameters’ button: configure display parameters.
- ‘Show SR’ button: show the x-y view of the reconstructed super-resolution image with each molecule color-coded by its axial position.
- ‘Export’ button: export the super-resolution image.

Note:

- After running 4Pi localization module or importing the 3D super-resolution reconstruction result, ‘Show SR’ button will be enabled, and ‘Export’ button will be disabled.
- After clicking ‘Show SR’ button, ‘Export’ button will be enabled.

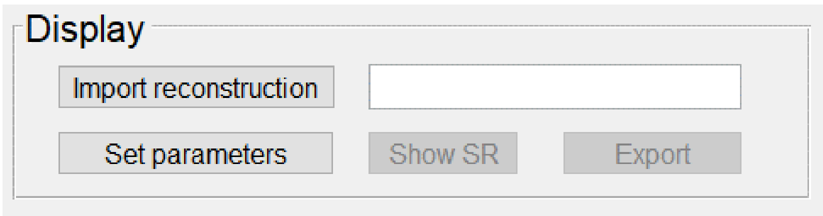

Figure 17. Display module interface.

The setting parameters in display module (Figure 18) include:

- SR image high bound: saturation intensity level of a reconstructed super-resolution image. Intensity above this bound will be set to saturation (255, in ‘tiff’ format).
- Zoom in: magnification in the reconstruction image. If we have a raw data frame with a size of 144 × 144 pixels and the ‘Zoom in’ value is set to 10, the reconstructed super-resolution image will have a size of 1440 × 1440 pixels.

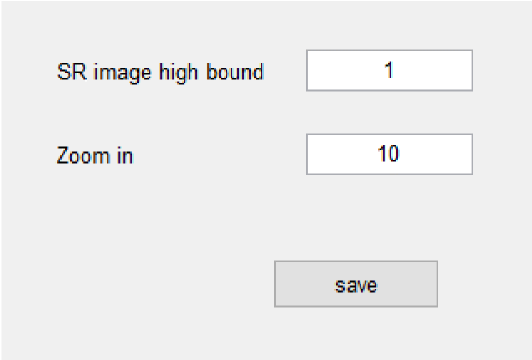

Figure 18. Setting parameters in display module.

The workflow in display module is as follows:

Step 1. Click ‘Set parameters’ button and configure the setting parameters.

Step 2. Click ‘Show SR’ button to show the super-resolution image.

Step 3. Click ‘Export’ button and export display results.

Note: If you have a previously generated 3D super-resolution reconstruction result, you can click ‘Import reconstruction’ button to import this file (e.g. ‘recon4Pi.mat’ file in ‘Data_4pi’ folder).

3. DATASET AND SOURCE CODES

3.1 Demonstration dataset

Data_4Pi\ Data_4Pi.mat	4Pi single-molecule dataset
Data_4Pi \ config_4Pi.mat	General setting parameters
Data_4Pi \ tform_all.mat	Alignment calibration file in 4 channels
Data_4Pi \ sCMOS_calibration_4Pi.mat	sCMOS calibration parameters
Data_4Pi \recon4Pi.mat	4Pi super-resolution reconstruction results

3.2 4Pi-BRAINSPOT source codes

'Main' folder

main.m	Main script for running 4Pi-BRAINSPOT
brainspot_4pi_GUI.m	Script for 4Pi-BRAINSPOT GUI
brainspot_4pi_GUI.fig	4Pi-BRAINSPOT GUI
default_cfg.mat	Default configuration for 4Pi-BRAINSPOT GUI
genPupilfigs_4ch.m	Script for generating figures of retrieved 4Pi-PSF model
export2csv.m	Script for exporting to 'csv' format
srhist_color.m	Script for generating color-coded super-resolution image

'Channel_alignment' folder

registration_4pi.m	Script for channel alignment in 4 channels
biplane_registration.m	Script for alignment between 2 channels

'Segmentation' folder

crop_subregion_4pi.m	Script for 4Pi-PSF segmentation
cMakeSubregions.mexw64	Mex function for cropping sub-regions

'4Pi_insitu_PSF_retrieval' folder

INSPR4Pi_model_generation.m	Script for estimating *in situ* 4Pi- PSF model
genIniguess.m	Script for estimating initial parameters
gen_init_4PiPupil_v2.m	Script for generating initial 4Pi coherent pupil
subregion_normalization_4pi.m	Script for normalizing sub-regions in 4 channels

gen_aber4PiPSF_directlyfromPR.m	Script for generating 4Pi-PSFs from coherent pupil
classify_4PiPSF_4channels_seperately.m	Script for classification and 2D alignment
registration_in_each_channel.m	Script for 2D alignment in each channel
dftregistration.m	Script for calculating sub-pixel 2D shift
FourierShift2D.m	Script for shifting image
cc2.m	Script for calculating 2D cross correlation
PR4Pi_fromAveZ_4channels_Zshift.m	Script for estimating 4Pi coherent pupil

'Dynamic_model_update' folder

analysis_cavityPhase.m	Script for estimating cavity phase
find_4pi_subregion.m	Script for 4Pi-PSF segmentation
genIniguess.m	Script for estimating initial parameters
caliphi0_NCC_v3.m	Script for estimating cavity phase using NCC
genini4pi_phi0_parfor_v3.m	Script for estimating cavity phase by incorporating additional variation into model
analysis_cavityPhase_and_misObj_v2.m	Script for estimating cavity phase and objective misaligned interferometric aberration
caliphi0_and_misObj_NCC_v3.m	Script for estimating cavity phase from NCC
genini4pi_misalign_parfor_v4.m	Script for estimating objective misalignment by incorporating additional variation into model
est_misobj_from_hist_v2.m	Script for estimating overall objective misalignment based on Gaussian fitting
est_phi0_from_hist_v2.m	Script for estimating overall cavity phase based on periodic-padding Gaussian fitting

'4Pi_localization' folder

analysis3D_from4pi_pupil_v2.m	Script for 4Pi reconstruction
crop_4pi_subregion_transModel_sCMOS.m	Script for 4Pi-PSF segmentation
loc_channel_specific_4Pimodel.m	Script for pupil-based 4Pi localization
genIniguess.m	Script for estimating initial parameters
cal_model_affine_4pi.m	Script for calculating affine matrix in 4Pi-PSF model
gensamplepsf_4pi_affine.m	Script for pre-generating 4Pi channel specific model
genpsf_4pi_real.m	Script for generating 4Pi-PSF model from pupil
genpsfstruct.m	Script for saving fitting parameters

genini4pi_z_parfor_affine.m	Script for estimating initial axial position
cuda4pi_loc_transModel.mexw64	Mex function for 3D localization
cuda4pi_sCMOS_loc_transModel.mexw64	Mex function for 3D localization with sCMOS calibration